H

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Practical Examination
Higher 2

CANDIDATE NAME	

BIOLOGY CLASS	2bi2_____	REGISTRATION NUMBER	

Biology 9744/04

Paper 4 Practical

16 August 2018

2 hours 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number on all the work you hand in.
Circle your practical shift and laboratory in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer questions one and two in the spaces provided on the Question Paper.
Answer question three on the Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your workings or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part of question.

Shift		
1	2	3
Laboratory		
BI23	BI24	CM43

For Examiner's Use	
1	**22**
2	**23**
3	**10**
Total	**55**

This document consists of **18** printed pages.

1 You are required to investigate the effect of varying the concentration of liver catalase on the rate of hydrogen peroxide decomposition. You will measure the rate of hydrogen peroxide decomposition using the apparatus shown in Fig. 1.1.

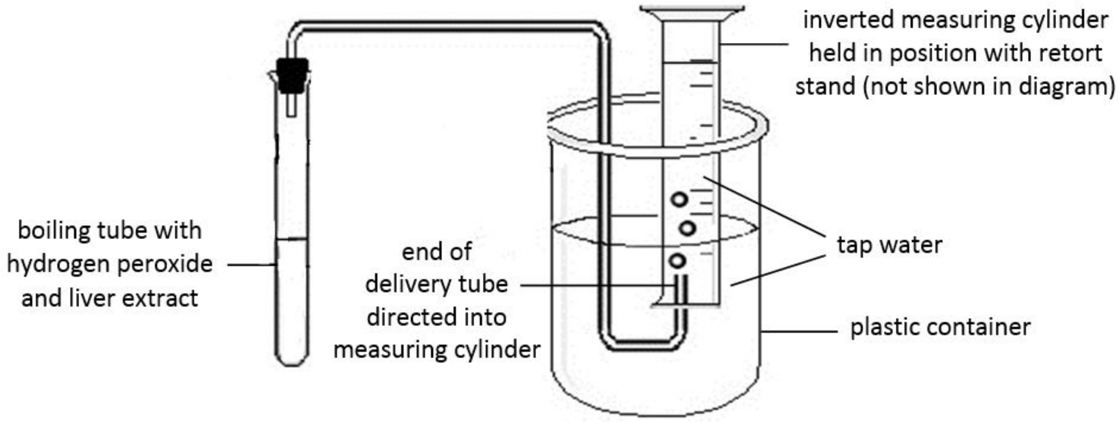


Fig. 1.1

Liver cells produce the enzyme catalase to catalyse the decomposition of the toxic chemical, hydrogen peroxide, to harmless products, water and oxygen gas.

$$2\ H_2O_2 \rightarrow 2\ H_2O + O_2$$

The oxygen gas produced by the liver extract in the boiling tube will be delivered to the inverted measuring cylinder, where it will displace some tap water.

You are required to:

- prepare a 100% liver extract
- perform a serial dilution to obtain different concentrations of liver extract
- determine the rate of hydrogen peroxide decomposition.

You are provided with:

- a small piece of chicken liver, in a sealable bag labelled **X**,
- 3% hydrogen peroxide solution, in a brown container labelled **Y**.

Hydrogen Peroxide is corrosive. Avoid contact with eyes or skin. If any should come into contact with your skin, wash immediately under cold water. Remember to wear safety goggles and latex gloves while performing this experiment.

*Before starting the investigation, read through steps **1** to **16** and prepare a table in **(a)(iii)**.*

Proceed as follows.

1 Press, with your fingers, on the small piece of chicken liver in the sealable bag labelled **X**, crushing it until it becomes a fine paste.

2 Mix the liver paste with 20 cm^3 of distilled water within **X**.

3 Line a 50 ml beaker with a filter bag.

4 Filter the liver preparation by transferring all of it from **X** into the filter bag with the help of a spatula. The filtrate obtained will be your 100% liver extract.

5 Prepare four other non-zero concentrations of liver extract by performing a serial dilution of your 100% liver extract with distilled water. You will need 1 cm^3 of each concentration for your experiment later.

(a) **(i)** Show clearly in the space below how you will prepare the four different concentrations of liver extract.

1. correct table headings and units
2. same dilution factor (e.g. 2) used for each step in the serial dilution
3. correct volumes (at least 1cm^3 prepared for each concentration)

concentration of liver extract / %	volume of liver extract taken from previous (higher) concentration / cm^3	volume of distilled water / cm^3
100	-	-
50	2.0	2.0
25	2.0	2.0
12.5	2.0	2.0
6.25	2.0	2.0

[3]

(ii) Explain why simple dilution is sometimes preferred over serial dilution.

- Simple dilution requires less stock solution to obtain the same concentration. It is thus preferred when stock solution is costly or limited in quantity.
- Simple dilution is done in a single step and thus has lower chance of transfer inaccuracies / does not require longer mixing times for better accuracy.
- Simple dilution allows concentrations with regular intervals to be obtained so that the gradual trend in data can be determined.

[1]

6 Fill a large plastic container with tap water to about four-fifth of its total volume.

7 Fill a 100 ml measuring cylinder to the brim with tap water.

8 Invert the measuring cylinder into the water in the large plastic container quickly. Some water may escape from the measuring cylinder, but there should be **at least 90 ml of water** left in the inverted measuring cylinder.

9 Use a retort stand to hold the measuring cylinder in position as shown in Fig. 1.1 such that the end of a delivery tube is directed into the measuring cylinder. The

mouth of the inverted measuring cylinder should remain below the water level in the large plastic container during this process.

10 Add 10 cm^3 of 3% hydrogen peroxide solution from **Y** into a clean boiling tube.

11 Add 1 cm^3 of the highest concentration of liver extract that you have prepared in step **5** into the same boiling tube using a small syringe. Shake the mixture well and quickly attach the delivery tube with a rubber bung to the boiling tube. Ensure that the connection between the rubber bung and the boiling tube is airtight.

12 Start the stopwatch immediately.

13 Allow the reaction to proceed for one minute and record the initial water level in the inverted measuring cylinder at the start of the second minute.

14 At the end of the third minute, quickly disconnect the rubber bung from the boiling tube and record the final water level in the inverted measuring cylinder.

15 Calculate and record the change in volume of water in the inverted measuring cylinder within the two minutes.

16 Repeat steps **6** to **15** using the other three concentrations of liver extract that you have prepared in step **5**.

(iii) Record your results in a suitable table in the space below.

Headings:

1. All headings are appropriate and with correct units.

Precision:

2. All volumes are recorded to the nearest 0.5cm^3 (i.e. 1 decimal place).

Trend:

3. Greatest change in volume of water in the inverted measuring cylinder within the two minutes is observed for the highest concentration of liver extract.

Content:

4. A total of 12 data for the four other non-zero concentrations of liver extract are recorded in the table.

concentration of liver extract / %	initial water level at the start of the second minute / cm^3	final water level at the end of the third minute / cm^3	change in volume of water within the two minutes / cm^3
50			
25			
12.5			
6.25			

[4]

(iv) Describe the observed effect of varying the concentration of liver catalase on the rate of hydrogen peroxide decomposition.

The higher the concentration of liver catalase, the faster the rate of hydrogen peroxide decomposition.

[1]

(v) Suggest a suitable control experiment that could have been used in this investigation.

Repeat the experiment using boiled and cooled liver extract.

[1]

(vi) Identify one variable that should be controlled in this investigation and describe how you could control it.

1. variable: temperature / pH
2. how to control: place boiling tube in thermostatically-controlled water bath / add pH buffer to liver extract

[2]

Turn over for the remainder of Question 1

(b) Some students studied the effect of varying pH on the rate of hydrogen peroxide decomposition by liver catalase using a different experiment setup as shown in Fig. 1.2.

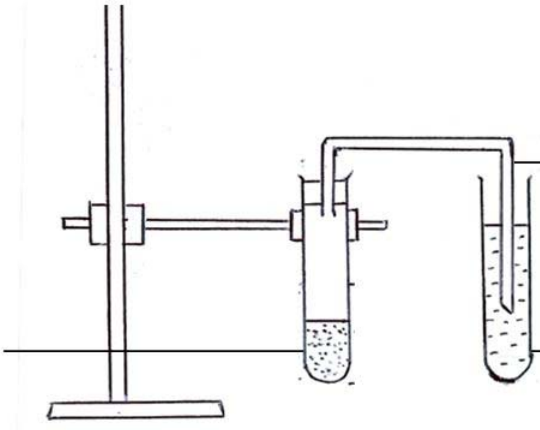


Fig. 1.2

The students' results are shown in Table 1.1.

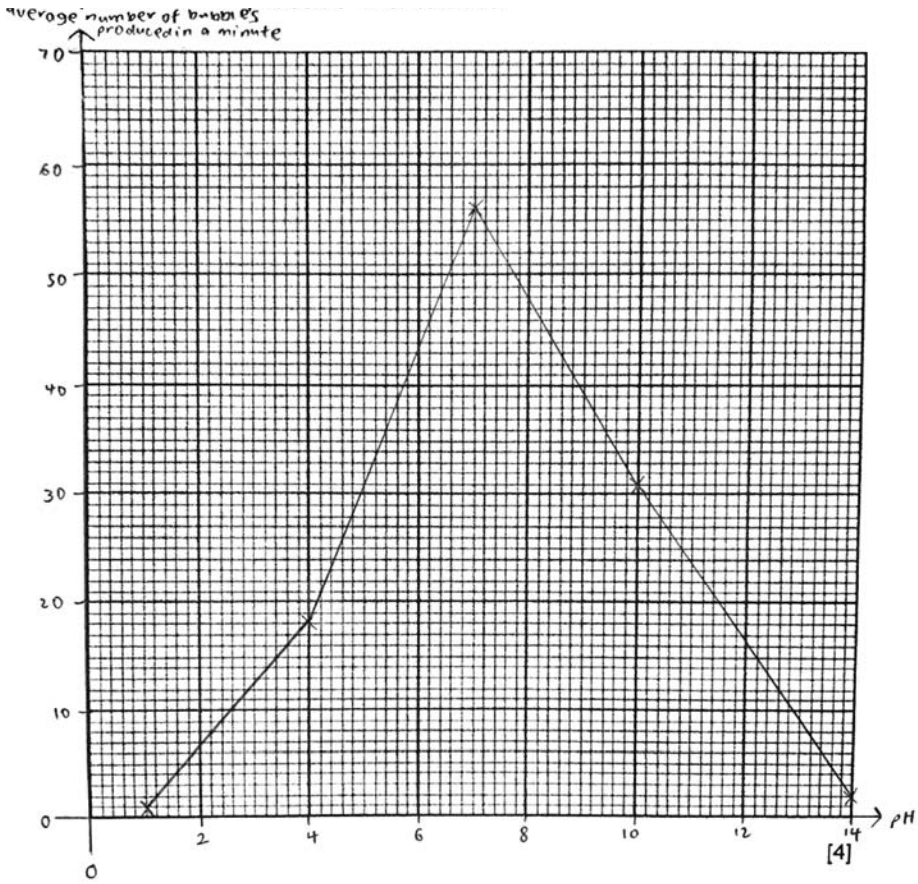
Table 1.1

pH	number of bubbles produced per minute			average number of bubbles produced per minute
	reading 1	reading 2	reading 3	
1	1	2	1	1
4	20	17	18	18
7	56	53	59	56
10	31	30	19	31 (reject 27)
14	2	3	1	2

(i) Calculate the average number of bubbles produced per minute for each pH and complete Table 1.1. [2]

- All five averages are calculated correctly.
  (anomaly excluded when calculating average for pH 10)
- All five averages are recorded to the nearest whole number.

(ii) Use the grid to show the observed effect of varying pH on the rate of hydrogen peroxide decomposition by liver catalase.



G – graph type (dot-to-dot plot? use ruler to draw lines?)

A – axis (orientation? labels? regular markings with consistent precision?)

P – points (all plotted? correctly plotted? no extrapolation?)

S – scale (sensible? at least half the grid?)

[4]

(iii) Explain the observed effect of varying pH on the rate of hydrogen peroxide decomposition by liver catalase.

1. pH 7 is nearest to the optimum pH as the average number of bubbles produced per minute is the highest at pH 7.

   OR

   As pH deviates (increases / decreases) from 7, the average number of bubbles produced per minute decreases.

2. It is only at the optimum pH that the R groups of the amino acids in catalase would acquire the appropriate charges and lead to the formation of the most ideal configuration that allows the enzyme to function most effectively.

   OR

   At non-optimum pH, the ionic charge of the acidic or basic groups of the amino acids in catalase would change and result in the disruption of ionic bonds that are involved in maintaining the enzyme's tertiary structure / active site. Hydrogen bonds in the enzyme structure would also be disrupted.

3. At optimum pH, hydrogen peroxide / substrate can bind to the active site of catalase / enzyme most effectively and be broken down into water and oxygen gas, thus leading to the highest average number of bubbles produced per minute.

   OR

   At non-optimum pH, hydrogen peroxide / substrate can no longer bind to the active site of catalase / enzyme as effectively and be broken down into water and oxygen gas, thus leading to lower average number of bubbles produced per minute.

[3]

(iv) Comment on the accuracy of determining the rate of hydrogen peroxide decomposition by counting number of bubbles produced per minute.

Counting bubbles produced per minute is less accurate than the water displacement method as the bubbles produced are of variable size / difficult to count the bubbles produced if the rate is too fast.

[1]

[Total: 22]

Question 2 starts on page 10

[Turn Over

2 During this question, you will require access to a microscope and a spectrophotometer.

S1 and **S2** are transverse sections of a banana at two different stages of development. You are required to examine the cellular structure of **S1** and **S2**.

Fig. 2.1 shows the cross-section of the banana used to prepare **S1** and **S2**.

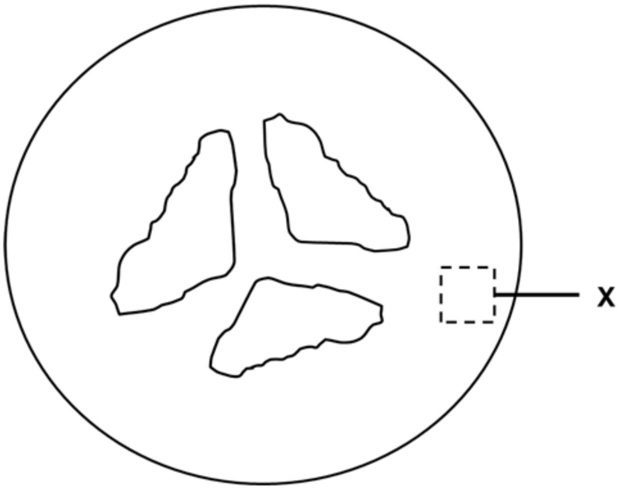


Fig. 2.1

(a) Cut a thin section, 3 mm by 3 mm by 1 mm, from region **X** of **S1** and place it on a clean microscope slide.

Add one drop of iodine to cover the section and leave for two minutes.

Place a cover slip over the sample. Place thumb over cover slip and press down firmly but gently as shown in Fig. 2.2.

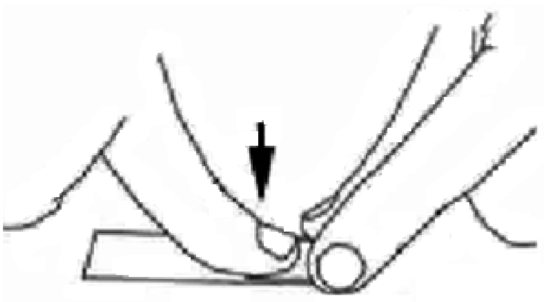


Fig. 2.2

(i) Examine the slide using a microscope. Observe the sample using all the objective lenses provided, and choose the lens that is more suitable for making observations of the cellular structure.

State which objective lens you have decided to use and give a reason for your choice.

40x + to view cellular organelles

[1]

(ii) State the number of eyepiece graticule units for one stage micrometer unit for the objective lens chosen in **(a)(i)**. Calculate the length of one eyepiece graticule unit in micrometer.

number of eyepiece graticule units 40

length of one eyepiece graticule unit 2.5µm

[2]

(iii) Make a large, labelled drawing of **three** adjacent cells in the space below. Indicate on your drawing the size of one cell in eyepiece graticule units.

C – clear, clean lines

S – size of three cells cover more than 2/3 space provided

P – proportion of cell wall to cell is accurate

L – shows labels cell wall, cell membrane, cytoplasm

[4]

(b) Repeat step **(a)** for **S2**.

Describe the observable differences between **S1** and **S2**.

S1 contained more intracellular structures stained blue-black/ starch grains

S1 cells were more elongated while S2 cells were more rounded

S1 cells were more regular in shape

S1 cells were packed closer together than S2 cells

[2]

*Before starting the investigation, read through steps **(c) - (i)** and prepare a table in **(f)**.*

(c) Cut a small section, 5 mm by 5 mm by 5 mm, from region **X** of **S1**, and place it in a plastic vial.

Add 10 cm^3 of distilled water into the plastic vial. Mash the section in water using a spatula. Leave the mixture to stand for five minutes.

Draw 2 cm^3 of the mixture from the top and place into a clean, dry test-tube. Add 2 cm^3 of Benedict's solution and place in a boiling water bath for three minutes.

Describe your observations.

Describes colour change (remained blue for S1)

[1]

(d) Transfer the mixture from **(c)** into a cuvette and measure the absorbance at **520 nm** using a spectrophotometer.

(e) Repeat steps **(c)** and **(d)** for **S2**.

(f) Record your absorbance results for **S1** and **S2** in a suitable table in the space below.

H – table shows column headings: specimen, absorbance/a.u.

R – absorbance reading recorded without units + S1<S2

[2]

(g) **Table 2.1** shows the absorbance readings of a series of glucose standards.

Table 2.1

concentration of glucose solution / %	absorbance
5.00	1.516
2.50	1.412
1.25	1.025
0.625	0.448
0.312	0.430

Use the grid to draw a best-fit graph to show the results in Table 2.1.

T – best fit graph plotted

A – x-axis: concentration of glucose solution/% + y-axis: absorbance/a.u.

S – no odd scale, graph spread covers more than ½ on each axis

P – data points are accurately plotted

[4]

(h) Show, on the graph plotted in **(g)**, how you will determine the concentration of glucose in **S1** and **S2**.

[1]

(i) Determine the concentration of glucose in **S1** and **S2** and record your results below.

concentration of glucose in **S1** ....

concentration of glucose in **S2** ....

[2]

(j) Comment on the observations of Benedict's Test and the absorbance results, and suggest an explanation.

1. Comments on agreement/discrepancy between Benedict's test and absorbance results
2. Provides logical explanation for point (1)
3. Makes logical inference about S1 and S2

[3]

(k) Suggest one limitation to your comment made in **(j)**.

Any logically reasoned limitation

[1]

[Total: 23]

Question 3 starts on page 16

3 Chlorophyll within a photosynthetic organism exists as pigment protein complexes in photosystem I (PSI) and photosystem II (PSII). Light energy absorbed by chlorophyll can:

(i) drive photosynthesis;

(ii) be radiated as heat; or

(iii) be re-emitted as light (fluorescence).

Measurement of the chlorophyll fluorescence is a tool to determine the stress level of a photosynthetic organism such as corals. This is done through the application of a saturating pulse of light (shown in Fig. 3.1(a)) using a Pulse-Amplitude Modulation (PAM) chlorophyll fluorometer (shown in Fig. 3.1(b)).

(a) **(b)**

Fig. 3.1

In a healthy, non-stressed organism which has been fully dark-adapted (i.e. kept in the dark for 12 hours), the value from the PAM chlorophyll fluorometer is highly consistent at approximately 0.83 arbitrary units. The existence of any type of ‘stress’ that results in inactivation damage of PSII will result in lower values.

As chlorophyll fluorescence is a measure of re-emitted light from PSII, naturally this means that any ambient light or pre-exposure of light to the corals before measurement can interfere with the measurement of fluorescence.

Design an experiment to determine whether increasing sea temperature would stress the coral species *Galaxea fasicularis*.

In your plan, you must use:

- *Galaxea fasicularis* fragments of about 5 - 7 cm radius, which has an optimal temperature for growth at 25 ^oC
- hand saw
- sterile sea water of 3.5% salinity
- plastic aquarium tanks
- aquarium heaters that will be placed into the tanks
- filtration and circulation pumps
- PAM chlorophyll fluorometer.

You may select from the following sterilised apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- timer, e.g. stopwatch.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 10]

Planning Mark Scheme

Pt	Type	Example	Mark Scheme:
1	Background	Increase in temperature, increase rate of photosynthesis by zooxanthellae in corals. Release of reactive oxidative species (ROS). Increase stress on corals and zooxanthellae. Results in inactivation damage of PSII in photosynthetic pigments in zooxanthella. Ref: protein structure in photosystem disrupted	Explain how temperature results in stress of coral tissues.
2	Hypothesis	Temperature above optimal causes inactivation damage of PSII, recorded value measured by using PAM chlorophyll fluorometer will decrease.	State hypothesis
3	Independent and Dependent Variables	Independent variable: temperature of sea water + 5 different values (with 1 value at approximately 25°C); (Maximum sea temperature is 35°C, accepted value is 40°C) Dependent variable: Recorded value from PAM chlorophyll fluorometer	State independent variable + 5 values uniformly spaced or derived. State dependent variable.
4	Variables to be controlled	Constant variables: fragment size of corals, species of corals, identical colony of corals, volume and concentration of sea water, amount of light, AVP (at least 2);	State 2 constant variables.
5	Rationale	State how one of the constant variable might affect the results.	State how one of the constant variable might affect the results.
6	Method	Prepare 5 plastic aquarium tanks with 2L of sea water at 3.5% salinity each	Describe how they are kept constant [CV4]
7	Method	Attached heater and filtration pump to the tanks and set individual tanks according to the temperature stated. Start both heater and filtration pump. **Tank** | **Temperature / °C** 1 | 28 2 | 30 3 | 32 4 | 34 5 | 36	Describe procedure to vary independent variable.
8	Method	Attach light source (60 W lamps) at a distance of approximately 10 cm away from the tank	Describe procedure to keep coral alive.
9	Method	Using the hand saw, saw out 15 coral fragments of approximately 3 cm by 3 cm	Explain why variables need to be kept constant + describe how they are kept constant [CV4]

<table>
<tr><td>10</td><td>Method</td><td>Place three coral fragment in each tank, and allow to acclimatise for 24 hours with the pre-set temperature</td><td>Describe procedure needed to ensure same starting point.</td></tr>
<tr><td>11</td><td>Method</td><td>After 24-hours, switch off the light and wrap the tank with aluminium foil for 12-hours, allowing the corals to dark-adapt.</td><td>Describe procedure for dark-adaptation.</td></tr>
<tr><td>12</td><td>Method</td><td>When dark-adaptation is completed, take 3 readings per coral to obtain a total of 9 readings per tank, by using PAM Chlorophyll Fluorometer and record the values in appropriate table.</td><td>Describe procedure to measure data.</td></tr>
<tr><td>13</td><td>Method</td><td>Well-labelled Diagram</td><td>Show diagram</td></tr>
<tr><td>14</td><td>Method</td><td>Describe control at 25°C / other method mark</td><td>Describe control set up</td></tr>
<tr><td>15</td><td>Reproducibility</td><td>Repeat entire experiment twice, using fresh solutions and coral fragments to ensure reproducibility of the results obtained.</td><td>Reliability and Reproducitbility</td></tr>
<tr><td>16</td><td>Results (Table)</td><td>
<table>
<tr><td rowspan="2">Temperature / °C</td><td rowspan="2">Coral Fragment</td><td colspan="4">PAM Reading</td></tr>
<tr><td>Reading 1</td><td>Reading 2</td><td>Reading 3</td><td>Average</td></tr>
<tr><td rowspan="3">28</td><td>1</td><td></td><td></td><td></td><td></td></tr>
<tr><td>2</td><td></td><td></td><td></td><td></td></tr>
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<tr><td rowspan="3">34</td><td>1</td><td></td><td></td><td></td><td></td></tr>
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<tr><td rowspan="3">36</td><td>1</td><td></td><td></td><td></td><td></td></tr>
<tr><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>3</td><td></td><td></td><td></td><td></td></tr>
</table>
</td><td>Show how results are to be presented in a table with independent and dependent variables in appropriate columns/rows [R1]</td></tr>
<tr><td>17</td><td>Results (Graph)</td><td>Graph to show trend;
Temperature on x-axis and average PAM reading on y-axis</td><td>Sketch graph to show relationship between independent and dependent variable.</td></tr>
<tr><td>18</td><td>Risks and Precautions</td><td>1. Ensure wires are properly insulated to prevent electrocution
2. Wear cotton/ladder gloves when handling the corals to prevent stings</td><td>Risks/safety</td></tr>
</table>